



Matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \end{bmatrix} \quad \begin{matrix} \text{column} \\ \text{1 row} \end{matrix}$$

$$B = \begin{bmatrix} 3 \\ 4 \\ -5 \end{bmatrix} \quad \begin{matrix} \text{1 column} \\ \text{3 rows} \end{matrix}$$

$$AB = [2(3) + 5(4) + (6)(-5)]$$

$$= [6 + 20 + (-30)]$$

$$= [-4]$$

$$3 \times 5 = 15$$

$$5 \times 3 = 15$$

$$1 \text{ row} \quad 1 \text{ column}$$

$$BA = \begin{bmatrix} 6 & 15 & 18 \\ 8 & 20 & 24 \\ -10 & -25 & -30 \end{bmatrix} \quad \begin{matrix} \text{3 rows} \\ \text{3 columns} \end{matrix}$$

$$1 \times 1 \quad AB = A \times B$$

$$3 \times 3 \quad BA = B \cdot A$$

$$A = \begin{bmatrix} 1 & 4 & -2 \\ 3 & 5 & -6 \end{bmatrix} \quad \begin{matrix} \text{2 rows} \\ \text{3 columns} \end{matrix}$$

$$B = \begin{bmatrix} 5 & 2 & 8 & -1 \\ 3 & 6 & 4 & 5 \\ -2 & 9 & 7 & -3 \end{bmatrix} \quad \begin{matrix} \text{3 rows} \\ \text{4 columns} \end{matrix}$$

$$AB = \begin{matrix} A & B \\ 2 \times 3 & 3 \times 4 \\ \text{same} & \\ 2 \times 4 \end{matrix}$$

$$BA = \begin{matrix} B & A \\ 3 \times 4 & 2 \times 3 \\ \text{X} & \end{matrix}$$

$$AB = 2 \times 4$$

BA = doesn't exist

$$AB = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 21 & 8 & 10 & 25 \\ 42 & -14 & 2 & 40 \end{bmatrix}$$

$$1 \times 1 = 1(5) + 4(5) + (-2)(-2) = 21$$

$$1 \times 2 = 1(2) + 4(6) + (-2)(9) = 8$$

$$1 \times 3 = 1(8) + 4(4) + (-2)(7) = 10$$

$$1 \times 4 = 1(-1) + 4(5) + (-2)(-3) = 25$$

$$2 \times 1 = 3(5) + 5(3) + (-6)(-2) = 42$$

$$2 \times 2 = 3(2) + 5(6) + (-6)(9) = -14$$

$$2 \times 3 = 3(8) + 5(4) + (-6)(7) = 2$$

$$2 \times 4 = 3(-1) + 5(5) + (-6)(-3) = 40$$

Modulus study

$a \equiv b \pmod{n}$ if and only if $n \mid a-b$ (n divides $a-b$)

a, b and n are integers; $n > 1$

Ex: $5 \equiv 2 \pmod{3}$

$3 \mid 5-2$

$a-b = nk$

$4 \equiv -1 \pmod{5}$

$5 \mid 4 - (-1)$

Ex. Find all possible values of n if $11 \equiv 5 \pmod{n}$

$n \mid 11-5$

$11 \equiv 5 \pmod{6}$



• Modular Arithmetic deals with remainders

• Divide a by n and let the quotient be q and the remainder r , then we have

$a = nq + r$ which implies $a \equiv r \pmod{n}$

Examples

13 divide by 5

$$\begin{array}{r} 2 \\ 5 \overline{)13} \\ \underline{10} \\ 3 \end{array}$$

quotient is 2 (q)

Remainder is 3 (r)

$13 \equiv 3 \pmod{5}$

$2 \equiv -3 \pmod{5}$

$5 \mid 2 - (-3)$

$-8 \equiv 7 \pmod{5}$

$5 \mid -8 - 7$

$5 \mid -15$

$-3 \equiv -8 \pmod{5}$

$5 \mid -8 - (-3)$

$5 \mid -5$

We can use addition, subtraction, and multiplication but division is problematic

Ex: Solve $x^2 + 3 \equiv 0 \pmod{7}$

$7 \equiv 0 \pmod{7}$

$x^2 \equiv -3 \pmod{7}$

$n \equiv 0 \pmod{n}$

$x^2 \equiv 4 \pmod{7}$

$\left\{ \begin{array}{l} x \equiv 2 \pmod{7} \\ x \equiv -2 \pmod{7} \end{array} \right.$

$nk \equiv 0 \pmod{n}$

$\{0, 1, 2, 3, 4, 5, 6\}$

Ex: $10 \equiv 4 \pmod{6}$

$6 \mid 10-4$

$10 \equiv 4 \pmod{6}$

$\downarrow$
 $5 \equiv 2 \pmod{6}$

Ex. $x^2 \equiv 2 \pmod{4} \rightarrow \{0, 1, 2, 3\}$
 $\downarrow \downarrow \downarrow \downarrow$
 $\{0, 1, 0, 1\} \leftarrow x^2$

No solution

What about negative exponents

$3x \equiv 1 \pmod{5}$

$\rightarrow 3x \equiv 6 \pmod{5}$

$3^{-1} \cdot 3x = 3^{-1} \cdot 1$

$x \equiv 2 \pmod{5}$

$x \equiv 3^{-1}$

$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$